CSC-612-001 Introduction to Computer Programming
Prof. Mroz
<u>Final Project</u>

Background: In machine learning and data science methodologies, one of the initial steps taken by an analyst is to explore the given data set to understand what it contains. This process is called Exploratory Data Analysis (EDA) and will be the first step students will take as they progress with this technology. The Final Project is a guided EDA assignment and needs to be completed and submitted as a Jupyter notebook. If the submission is done on Google Colab, **<u>the link MUST be provided in Moodle, not via email.</u>**

Data Set (from Kaggle): "Customer Satisfaction in Airline." The data for this activity is for a sample size of 129,880 customers. It includes data points such as class, flight distance, and in-flight entertainment, among others. – The intended use of this data set is to understand the various factors that influence an airline traveler's satisfaction in their journey. Ultimately this analysis can provide a roadmap for predicting if a customer would be satisfied.

Instructions: Please answer the following questions by writing the python code to meet the requirements of each question. Utilize a single Jupyter notebook file and name it in the format of "lastname_firstname_FINAL.ipynb". Please utilize multiple cells as necessary and remember to utilize markdown cells, comments, header, and internal documentation.

<u>The file must be executable by the professor for full credit as well as it must adhere to the rubric and coding standards.</u>

1. The data set is contained within the file "Airline.csv" which is provided on Moodle which needs to be downloaded and saved in a directory which is accessible by your notebook. Load the csv file into a pandas dataframe and display the top 5 lines on the screen to validate it was loaded correctly. **Take note that the first row in the csv file contains the column names (features)

2. The first analysis needs to be done on the features of the data which identify the customer types. Looking at the data, **identify which features can be used to classify the different types of passengers**. Create a new dataframe for ***<u>each classification feature</u>*** that you identify. Print to the screen the first 5 rows of each dataframe to validate their creation. *For example, dataframes need to be created for customers classified by the "satisfaction" feature where one is "satisfied" and the other is "dissatisfied". **There must be one data frame for EVERY identified feature. Only creating a "satisfaction" dataframe is not sufficient and will impact the ability to answer the following questions correctly.**

3. In order to learn more about the customer demographic, create a histogram of the "Age" for **each customer classification dataframe** and print them to the screen. This could show if certain age groups fall into different categories.

4. This is a single airline company's data set, so flights of the same distance would be considered a single "route". To determine the number of specific routes this airline has, create a frequency dictionary of the **aggregated dataframe (includes all features)** on the "Flight Distance" feature. Print to the screen the dictionary for verification. Then print to the screen the total number of distinct "routes." *Please note that "Flight Distance" is numeric, but you will need to treat it as categorical.*

5. This data set is an aggregated view from a customer survey. For **each of the customer classification dataframes**, calculate the average score of each feature. The last two features measure the time in minutes of arrival and departure delay. Please calculate those averages as well for each dataframe. **Save each dataframe's averages into lists specific to each dataframe and print the lists to the screen for validation.**

6. Considering the **"satisfied" and "dissatisfied"** dataframes (which should only have satisfied or dissatisfied customers respectively), how many *total satisfied customer*s are there compared to the *total of dissatisfied*? Which "Type of Travel" is most frequent in the satisfied dataframe versus dissatisfied? Which "Class" is the most frequent in the satisfied versus dissatisfied dataframe.

7. Considering the code written above and the results of the data outputs, please write in English (not-code) **in a markdown cell** a short answer response to the question:

   In your opinion, which feature is most important for the Airline to consider when they are working to increase customer satisfaction? Why did you choose that feature, and what did the data show to support your reason? (3-4 sentences max.). ***For this answer to be considered correct, the analysis must refer to data that was generated with the code submitted.***